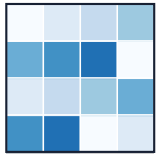
One trained ViT, two attribution readouts

Different patches can have different scores; uniform expansion acts inside each patch.

a The model represents structure within and between patches



224 × 224 image



16 × 16 × 3 patch

Patch embedding



A feature vector per token



Pixel positions can contribute differently.

Self-attention

Weights vary across tokens,

inputs, blocks and heads.

Select one internal target

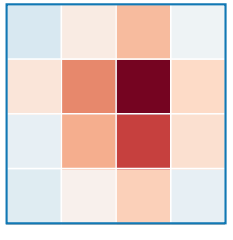
$$f(z) = u^\top g(z)$$

Same target for both methods

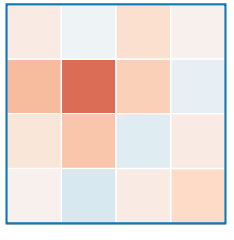
b NNS: contributions in input coordinates

Within-patch detail is computed from input gradients.

Input patch A



Input patch B



16 fine cells

per patch

Each cell: 4 × 4 pixels

Different input locations may receive different signed scores.



negative ← 0 → positive

c Matched token baseline

Patch baseline: one token score → uniform cells in its patch

Token A: cA = 16

1	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

Patch A: 1 per cell

Token B: cB = 8

0.5	0.5	0.5	0.5
0.5	0.5	0.5	0.5
0.5	0.5	0.5	0.5
0.5	0.5	0.5	0.5

Patch B: 0.5 per cell

A ≠ B

Different between patches

Equal within each patch

Uniformity comes from this readout rule.

Illustrative grids and scores. The two readouts are computed independently; patch-wise totals are not constrained to match.